EE236: Electronic Devices Lab
Lab 4: I/V Characteristics of Solar Cell

Abhineet Agarwal, 22B1219

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1 Aim of the experiment

The aim was to perform the following tasks for a solar cell:

- Measurement of I-V characteristics in forward and reverse bias of the solar cell in the dark.
- I-V characteristics for two different levels of illumination I1 and I2.
- Measurement of VOC and ISC at different illumination levels.

2 Theory

A solar cell is just a large area pnjunction. The LEDs used in the previous experiments are also pn junctions. The most noticeable difference between these two types of pn junctions is the area of the device. The area of a LED is small compared to the area of a solar cell. A typical industrial solar cell is 15cm x 15 cm pn junction. In contrast a LED is typically 2mm x 2mm pn junction.

Principle of operation: We first examine what happens when light is incident on a semiconductor. Light of energy (hν) greater than the band gap E_g is absorbed. When a photon is absorbed, an electron from the valence band is excited into the conduction band. Each absorbed photon gives rise to a free electron and hole.

In the dark in forward bias, electrons from the n side are injected into the p side and similarly for holes. In the dark, the diffusion component of the current increases exponentially with voltage (this is what you saw in previous lab). Under illumination, excess minority carriers are created in the semiconductor. Again, for simplicity the majority carriers are not shown (remember that electrons are minority carriers in p side, and holes are minority carriers in n side). The excess minority carriers are separated by the pn junction and electrons travel to the n side (and holes to the p side), giving rise to current, called as photocurrent. The direction of this photocurrent is opposite to the diffusion current in forward bias. Thus, the total current (I_{total}) flowing in the diode in forward bias when voltage (V) is applied, can be expressed as:

$$I = I_{total} = I_{diffusion} - I_L = I_0[e^{qV/\eta kT} - 1] - I_L \tag{1}$$

where, η is the ideality factor of the diode, I0 is the saturation current and IL is the photocurrent. Thus, when the illumination is increased, more photons are absorbed and the magnitude of IL increases. Due to the difference in sign, this manifests as a down-shift in the I-V curve.

Given the way we have defined the direction of current flow, positive sign for I indicates that the diode sinks current. Negative sign for I indicates that the diode sources current i.e. it acts as a power supply. This is the regime of operation that we are interested in (we want to generate power with a solar cell). Thus, the 4th quadrant of this I-V or P-V curve is the one we are interested in. The y-intercept on the I-V curve (i.e. V=0, or short) is called short-circuit current (I_{SC}), and the x-intercept (i.e. I=0, or open) is called open-circuit voltage (V_{OC}). The power at I_{SC} and V_{OC} operation is zero, and maximum power can be obtained by choosing a voltage equal to the value where the yellow rectangle touches the I-V curve. These corresponding values for maximum power are denoted I_{MP} and V_{MP}. The fill factor is defined as ratio of the area under the rectangle ($V_{MP} \times I_{MP}$) to the product ($V_{OC} \times I_{SC}$), thereby telling us how much power the solar cell can really generate as a fraction of the area under the I-V curve in the 4th quadrant.

3 Experiment

3.1 Measurement of I-V characteristics

3.1.1 Circuit

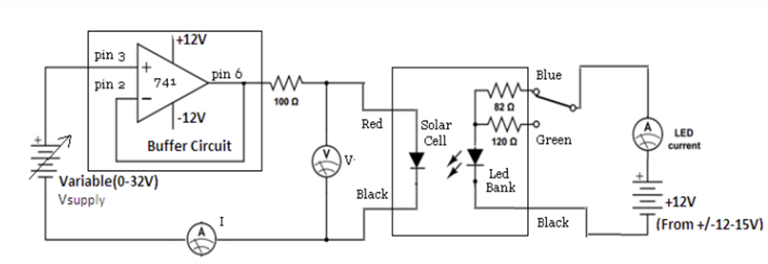


Figure 1: I-V characteristics circuit

3.1.2 Readings

Part-A : Dark I-V characteristics

Table 1: Voltage vs. Current Data

Vd (V)	Id (mA)
-1.9	-1.08
-1.72	-0.81
-1.54	-0.63
-1.35	-0.47
-1.16	-0.35
-0.97	-0.27
-0.78	-0.21
-0.58	-0.16
-0.39	-0.13
-0.24	-0.1
-0.09	-0.06
0	0
0.09	0.05
0.17	0.19
0.31	0.79
0.38	1.93
0.42	3.44
0.43	4.26
0.44	4.67
0.45	5.95
0.46	7.25
0.47	9.46
0.48	11.68
0.49	13.9

Part-B

a) I_1: Green

- I_{LED}: 41.4 mA

Table 2: I-V Data for Solar Cell under illumination I1

Voltage (V)	Current (Id) (mA)
0.5	11.96
0.49	6.53
0.48	2.94
0.47	1.17
0.46	-0.59
0.45	-2.32
0.44	-3.17
0.43	-4.02
0.42	-4.82
0.4	-5.58
0.34	-6.89
0.22	-7.69
0.07	-8.08
-0.11	-8.19
-0.31	-8.26
-0.5	-8.3
-0.7	-8.35
-0.89	-8.43
-1.08	-8.51

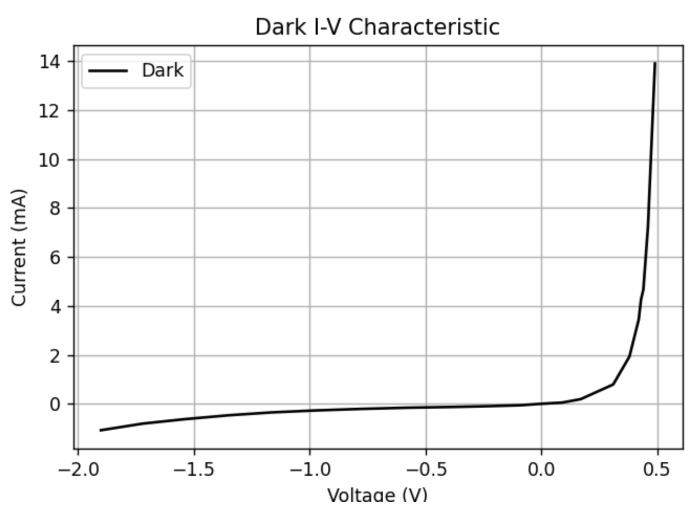
b) I_2: Blue LED

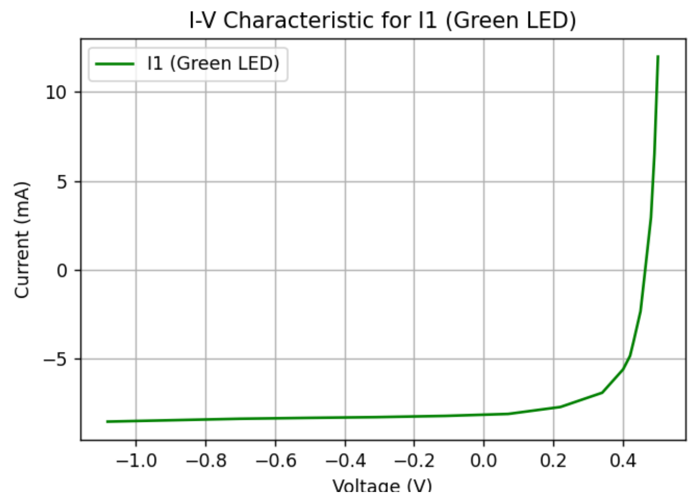
- I_{LED}: 58.5 mA

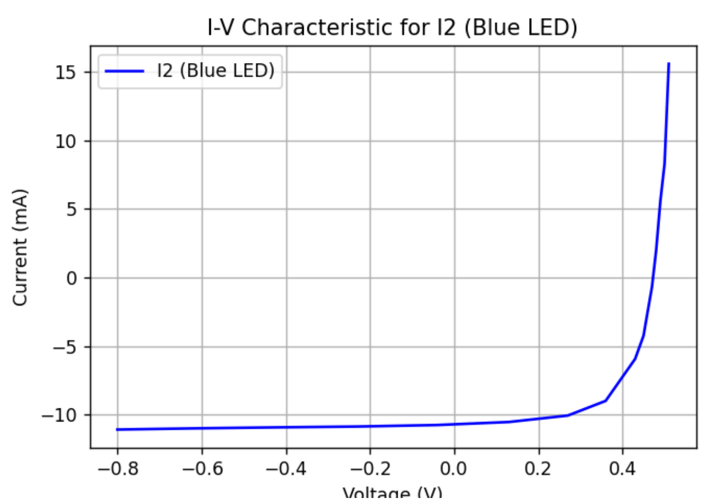
Table 3: I-V Data for Solar Cell under illumination I2

Voltage (V)	Current (Id) (A)
-0.8	-11.09
-0.61	-11
-0.42	-10.93
-0.23	-10.87
-0.04	-10.76
0.13	-10.54
0.27	-10.07
0.36	-9
0.43	-5.94
0.45	-4.25
0.46	-2.46
0.47	-0.7
0.48	1.98
0.49	5.57
0.5	8.29
0.51	15.58

3.1.3 Results and Inferences




I-V Characteristic for I1 (Green LED)
I1 (Green LED)
Current (mA)
10
5
0
−5
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
Voltage (V)


I-V Characteristic for I2 (Blue LED)
I2 (Blue LED)
Current (mA)
15
10
5
0
−5
−10
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
Voltage (V)

3.2 Solar cell as power source

3.2.1 Circuit

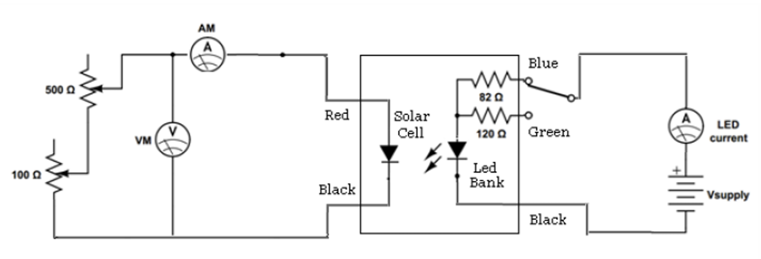


Figure 2: Circuit for Solar cell as power source

3.2.2 Readings

Illumination 1

- I_{LED}: 41.2 mA

V(d) (V)	I(d) (A)
0.085	7.98
0.112	7.94
0.138	7.90
0.163	7.84
0.188	7.76
0.230	7.63
0.257	7.50
0.291	7.30
0.345	6.75
0.370	6.43
0.387	6.00
0.400	5.59
0.411	5.15
0.422	4.69
0.436	3.66
0.446	2.77
0.452	2.13
0.459	1.22
0.461	0.91
0.462	0.70

Table 4: I-V Data for illumination 1

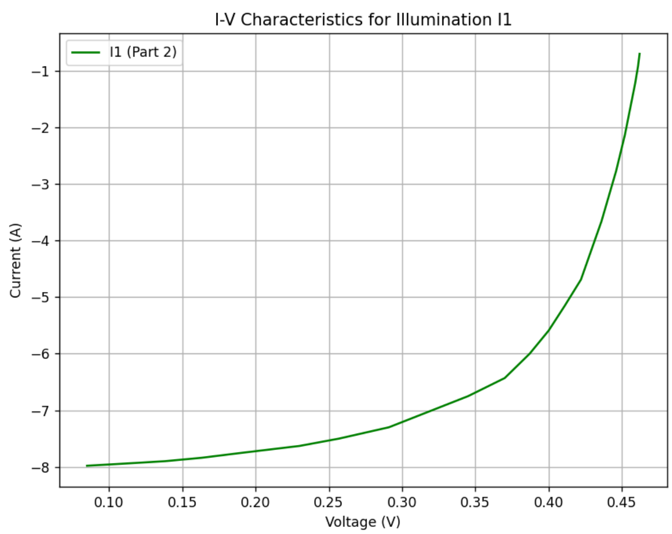
Illumination 2

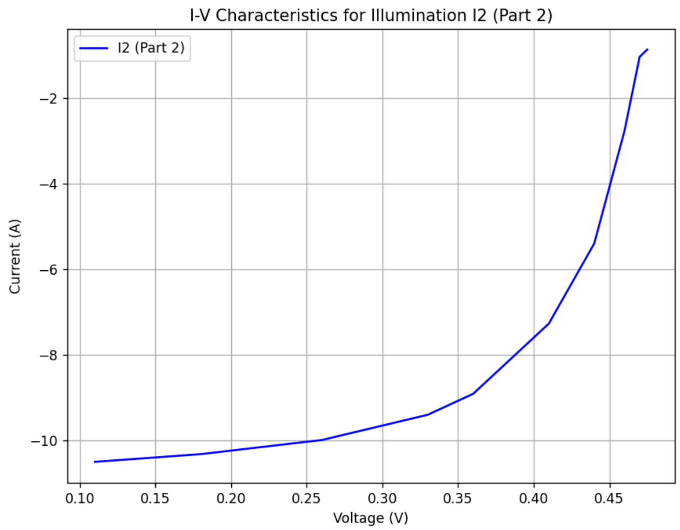
- I_{LED}: 54.5 mA

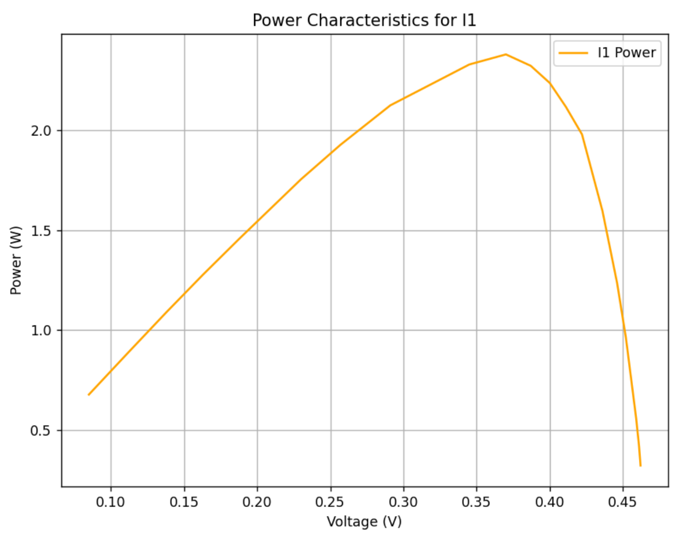
Voltage (V)	Current (A)
0.11	10.5
0.18	10.32
0.26	9.99
0.33	9.40
0.36	8.91
0.41	7.27
0.44	5.40
0.46	2.77
0.47	1.04
0.475	0.87

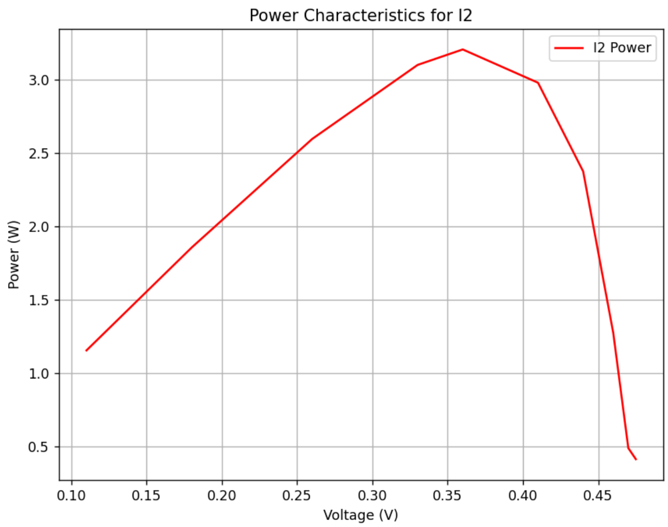
Table 5: I-V data for illumination 2

3.2.3 Results and Inferences



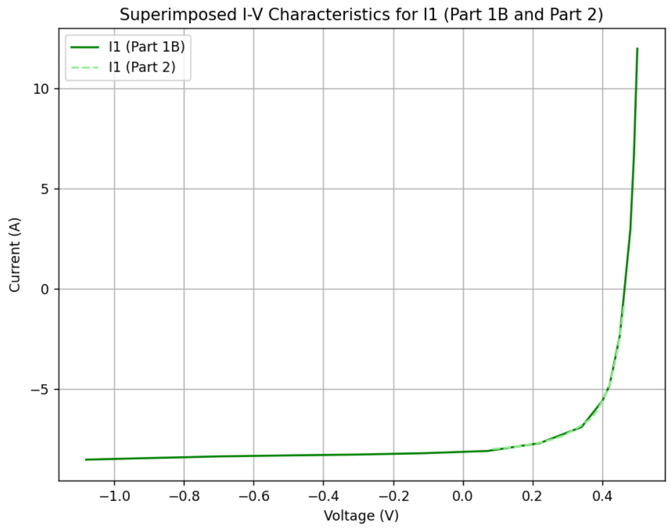

I-V Characteristics for Illumination I2 (Part 2)
I2 (Part 2)
Current (A)
Voltage (V)
−2
−4
−6
−8
−10
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45

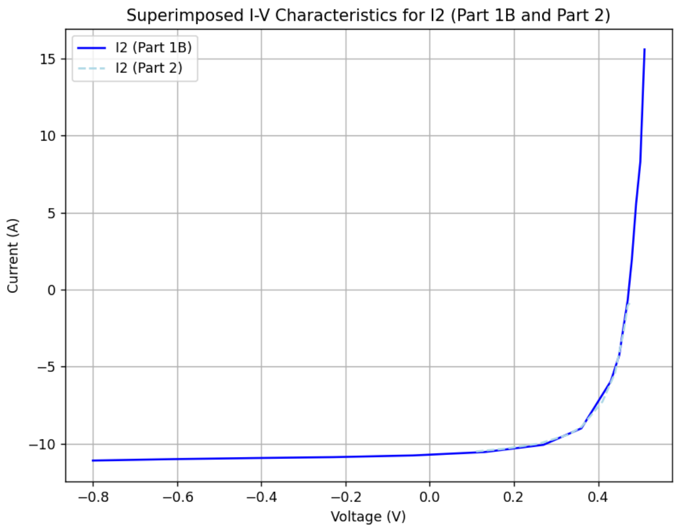

Power Characteristics for I1
I1 Power
Power (W)
Voltage (V)
2.0
1.5
1.0
0.5
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45



Parameter	**I1**	**I2**
Isc (A)	-8.106	-10.783
Voc (V)	0.465	0.501
VMP (V)	0.37	0.36
IMP (A)	-6.43	-8.91
Fill Factor	0.631	0.594

Table 6: Summary of I-V Parameters for I1 and I2

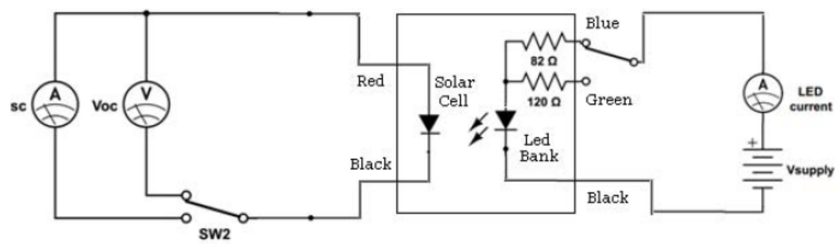




As can be seen, the two curves do match in the fourth quadrant for both the cases.

3.3 Measurement of V_{OC} and I_{SC} at different illumination levels

3.3.1 Circuit

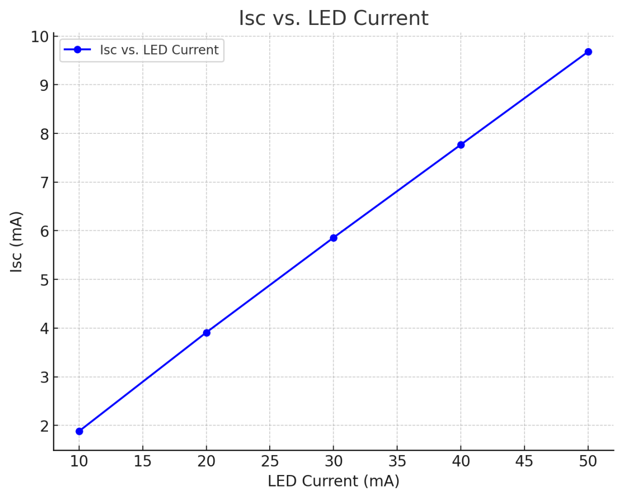


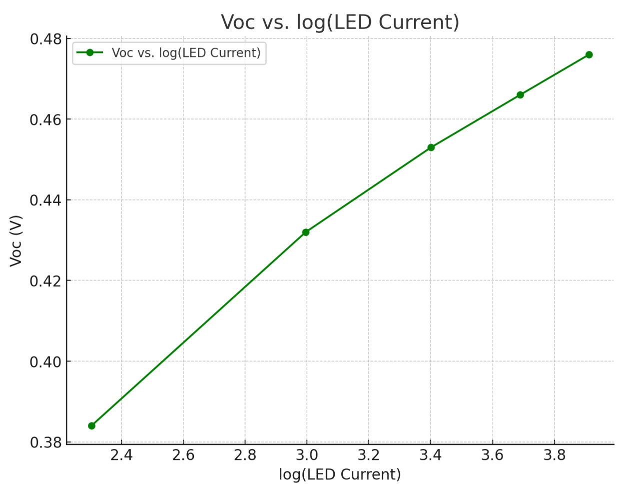
3.3.2 Readings

LED current (mA)	V_{OC} **(V)**	I_{SC} **(mA)**
10	0.384	1.88
20	0.432	3.91
30	0.453	5.86
40	0.466	7.77
50	0.476	9.68

Table 7: LED current vs. V_{OC} and I_{SC}

3.3.3 Results and Inferences





This experiment shows that I_{SC} varies linearly with light intensity and V_{OC} varies linearly with log of intensity.